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$E = \omega^2 R^3$: Kepler's Third Law as the Basis of a Unified Physical Picture

Results of the Ew2R3.org Programme and a Call for Independent Verification

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Abstract

We propose that Kepler's third law in the form $E = \omega^2 R^3$ expresses one physical process: pressure-driven motion of a non-material medium towards aggregates of impermeable atoms. The medium flows around atoms and, in the regimes considered, has the properties of an ideal incompressible fluid/gas with zero internal viscosity. We derive the acceleration of a specified radial flow, controlled grouping bounds, transition and compatibility conditions, and a conditional common response for weight and inertia. The formula connects these results across the series. In the central regimes examined, the baseline predictions coincide with the standard description at $E = GM$. The test-body mass cancels from the force-to-acceleration relation, the kilogram cancels in the dimensions of GM, and an orbit determines E directly from lengths and periods, without G or the source mass M. We give a numbered register of concrete contributions and a programme of independent checks. We call for verification of the stated calculations and physical premises, and research support for quantitative discriminating predictions.

Keywords: Kepler's third law, orbital parameter, non-material medium, multiple systems, weight and inertia, independent verification.

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1 The formula as the organizing principle

1.1 The meaning of the worldview formulation

Kepler's third law is known and used in celestial mechanics. Our thesis concerns the broader physical role of its angular-velocity formulation. We propose using $E = \omega^2 R^3$ as a common starting parameter for motion, acceleration and interaction with a medium, beyond the relation between orbital periods and sizes. We argue that this interpretation deserves substantially wider investigation. This is the position of our programme, not a measured assessment of scientific awareness.

The first studies establish the starting relations and develop the radial interpretation [1, 2, 3]. The companion studies examine cumulative motion, multiple systems, transitions between regimes, body–flow coupling and flow around atoms [4, 5, 6, 7, 8]. This synthesis connects their results and identifies a common subject for independent verification.

1.2 Orbital definition and dimensions

For central circular motion of radius R and period T,

$$\omega = \frac{2\pi}{T}, \qquad E = \omega^2 R^3 = v^2 R, \qquad g = \frac{E}{R^2}. \tag{1}$$

Here g is the required centripetal acceleration. Extending this relation to other radii adopts an inverse-square central law. For a Kepler ellipse the correct invariant is $E = n^2 a^3$, with $n = 2\pi/T$ and semimajor axis a. Instantaneous $\omega^2 r^3$ is generally not constant on an ellipse.

$$[E] = L^3 T^{-2}, \quad \frac{E}{R^3} = \omega^2, \quad \frac{E}{R^2} = g, \quad \frac{E}{R} = v^2. \tag{2}$$

The last three quantities have different dimensions: squared frequency, acceleration and squared speed. E is not a potential. The associated central potential per unit inertia is $\Phi = -E/r$. Nor is E one number shared by all bodies. A planet's orbit determines the parameter of the Sun or of the corresponding relative system. A planet's own parameter requires satellite data or another appropriate dynamical determination.

The operational calculation does not require separate numerical inputs for M and G. This agrees with the use of directly determined GM parameters in JPL ephemerides [11]. Eliminating separate inputs is not evidence for a new mechanism. Our contribution concerns the physical interpretation and testable connections between the tasks below.

1.3 What cancels in the standard mass notation

For a source of mass M, the Newtonian comparison distinguishes the gravitational and inertial masses of a test body,

$$F = \frac{GMm_{\mathrm{g}}}{r^2}, \qquad a = \frac{F}{m_{\mathrm{i}}} = \frac{GM}{r^2}\frac{m_{\mathrm{g}}}{m_{\mathrm{i}}}. \tag{3}$$

At $m_{\mathrm{g}} = m_{\mathrm{i}}$, the test-body mass cancels exactly. There is also a dimensional cancellation,

$$[GM] = (\mathrm{m^3\,kg^{-1}\,s^{-2}})(\mathrm{kg}) = \mathrm{m^3\,s^{-2}} = [E]. \tag{4}$$

This cancels the kilogram unit in the product, not the source parameter algebraically. Operationally, $E = 4\pi^2 a^3/T^2$ is determined from orbital lengths and time, without separately supplying G and M. For a relative binary orbit it measures the pair parameter, not one component's parameter. In the standard theory that parameter is $G(M_1 + M_2)$. Thus $g = E/r^2$ is a calculation without separate mass inputs, while the physical explanation of its universality is the subject of the coupling model below. The dimensions alone do not determine that explanation.

2 Numbered register of the series contributions

This register states the contributions for which we claim authorship: the combined physical interpretation, the specified constructions and their tests. Classical Keplerian and hydrodynamic formulas retain their attribution. The equations identify the assumptions under which each result is established.

1. **One orbital parameter, distinct source meanings.** We connect orbital lengths and periods to E, central acceleration and characteristic speeds without separate numerical mass inputs. We distinguish a source parameter, a relative-pair parameter and a group parameter [1, 2, 5]. The test-body and dimensional cancellations are given in Eqs. (3)–(4).

2. **Where two prescribed radial regimes meet.** For a cumulative moving front, $R^3\dot{R}^2 = C = E\ell^2$, its acceleration equals $-E/R^2$ at $R_* = \ell\sqrt{3/2}$. Equality of front and released-body speeds at the surface gives $R_0 = R_b/[1-\ell^2/(2R_b^2)]$, finite for $\ell^2 < 2R_b^2$. The threshold is $2R_b$ specifically when $\ell = R_b$ [4]. These compare front and body kinematics. Identification of the front speed with a local medium speed requires a spatial flow prescription.

3. **An all-direction source-group bound.** For positive source parameters in the specified additive law, let $E_G = \sum E_j$, $\mathbf{c}_G = \sum E_j\mathbf{r}_j/E_G$, $a_G = \max|\mathbf{r}_j - \mathbf{c}_G|$, and $Q_G = \sum E_j|\mathbf{r}_j - \mathbf{c}_G|^2/E_G$. Then

$$\frac{|\mathbf{a}_G - \mathbf{a}_{\text{mono}}|}{E_G/R^2} \le \frac{3Q_GR^2}{(R-a_G)^4}, \qquad R > a_G. \tag{5}$$

   Here $R = |\mathbf{x} - \mathbf{c}_G|$, $Q_G = m_2$ and $a_G = b$ in the companion notation [5]. For $F(\mathbf{x}) = \mathbf{x}/|\mathbf{x}|^3$, the Hessian bound $\|D^2F[\mathbf{h},\mathbf{h}]\| \le 6|\mathbf{h}|^2/|\mathbf{x}|^4$ and the Taylor factor $1/2$ give coefficient 3. For a pair of unit separation with $E_1 = 0.75$, $E_2 = 0.25$, and $E_3 = 0.3$ at $D = 20$, the sufficient initial interval is $4.735314 \le R \le 6.078802$. Internal grouping and external differential acceleration are each at most 5% against the same monopole scale. Their sum is at most 10%. This is a spatial field bound, not a stability theorem.

4. **A tidal approximation checked against an exact prescribed field.** For TIC 454140642, the combined leading tidal approximation has a largest sampled relative discrepancy 2.594824% over 3457 prescribed states spanning 432.1 days [5]. The reference is the exact external differential acceleration of those states. This is a model-comparison result, not observational accuracy or an all-time bound.

5. **Corotation without individual radius fits.** From each body's E and spin Ω, we calculate $R_\Omega = (E/\Omega^2)^{1/3}$ for ten bodies [6]. The Moon and Phobos lie on opposite sides of the corresponding corotation radius, consistent with their opposite secular drift directions. This sign consistency is also present in conventional tidal descriptions and does not discriminate between mechanisms.

6. **A three-dimensional continuity construction.** We construct a divergence-free axisymmetric field with a cubic radial contribution and meridional transport,

$$\begin{aligned} u_r &= -A/r^2 - Br^{-3/2}P_2(\cos\theta), \\ u_\theta &= (B/4)r^{-3/2}\sin\theta\cos\theta, \\ u_\phi &= r\Omega(r)\sin\theta, \end{aligned} \tag{6}$$

   where $P_2(x) = (3x^2-1)/2$. The coefficient $1/4$ cancels the radial divergence. Inflow at every latitude requires $-1 < B\sqrt{r}/A < 2$ for $A > 0$. This field generally has vorticity. Its total radial $r^3u_r^2$ is not a global constant when $A \neq 0$ [6]. For the prescribed HD 142527 profile, $C_r \propto r^2(L-r)^7$, $L = 143$ au, the maximum is at $2L/9 = 31.7778$ au. On 30–35 au, its departure from that maximum is at most 1.256%. This is local flatness of a model profile, not accuracy against a measured medium speed.

7. **An exact passive compatibility criterion and a boundary obstruction.** Set $\mathbf{w}_0 = \mathbf{v}_0 - \mathbf{u}$ and $\mathbf{d} = \mathbf{a}_m - \mathbf{g}$. For $\mathbf{w}_0 \neq 0$, a symmetric positive-semidefinite K satisfying $K\mathbf{w}_0 = \mathbf{d}$ exists exactly when $\mathbf{w}_0 \cdot \mathbf{d} > 0$ or $\mathbf{d} = 0$ [7]. If $\mathbf{w}_0 = 0$, compatibility requires $\mathbf{d} = 0$. At the fixed-normal, all-inward outer equator, the prescribed corotation condition instead gives $\mathbf{w}_0 \cdot \mathbf{d} = -U^2 D < 0$, with $U = u_r < 0$ and $D = \partial_r u_r > 0$. This excludes that boundary class, not all interactions. The matrix criterion is an existence test, not an independently derived constitutive law.

8. **A simultaneous comotion and free-fall condition.** In the nondissipative class $\dot{\mathbf{v}} = \kappa D_t \mathbf{u}$, for a steady inward radial field vanishing at infinity, invariance of comotion and the central release target together require $\kappa = 1$ and $U^2 r = 2E$ [7]. The local pressure–added-inertia comparison gives $\kappa = (1 + c_a)/(\beta + c_a)$, hence unit response at $\beta = 1$ [4, 8]. Different directions of body motion are retained in the coupling tests.

9. **Acceleration, rates and the body's approach to flow speed.** For $U = \sqrt{2E/r}$ and $\lambda_{\mathrm{m}} = \sqrt{E/(2r^3)}$, we derive

$$E/r^3 = 2\lambda_{\mathrm{m}}^2, \qquad E/r^2 = \lambda_{\mathrm{m}} U, \qquad E/r = U^2/2. \tag{7}$$

   We evaluate these model quantities for ten bodies [8]. A body released from rest has $|v_r/u_r| = \sqrt{1 - r/r_0}$ at unit response and no velocity-dependent damping. The squared rate, acceleration and squared-speed scale are distinct quantities. λ_{m} is a flow-gradient rate, not the additional interaction coefficient λ of the gravimetric test.

10. **Conditional acceleration additivity.** A smooth field $\mathbf{u} = -\nabla\psi$ satisfying

$$-\partial_t \psi + \tfrac{1}{2}|\nabla\psi|^2 = W + C(t), \qquad W = \sum_j E_j/r_j \tag{8}$$

    obeys $D_t \mathbf{u} = \nabla W$ exactly [8]. The additive acceleration follows from the imposed Hamilton–Jacobi equation, not from directly adding individual radial velocities. Moving sources require a time-dependent solution.

11. **Reproducible calculations and a separate instrument test.** The software connects formulas, frozen inputs, synthetic controls and interactive configurations [9, 10]. The conditional gravimetric contrast in Eq. (13) specifies an additional test. Software releases document implementation, while physical confirmation requires independent measurement of the proposed medium response.

3 The physical picture and its mathematical conditions

3.1 Flow around impermeable atoms

We propose that atoms are impermeable cavities formed by collapsed three-dimensional standing waves. Atomic aggregates create pressure differences. Under its own pressure, the non-material medium moves from higher to lower pressure and flows around atoms, including accessible exterior channels and an open toroidal hole. It does not enter the closed interior cavity of an existing atom. Within the stated regimes the medium has the properties of an ideal incompressible fluid/gas [4, 7, 8].

Narrowing of channels can increase the local flow speed. The resultant force is determined by the boundary pressure integral. An increase in speed alone does not establish nonzero drag. Internal viscosity, medium–atom interaction and atom–atom interaction are distinct questions. This article does not establish a connection between the medium and the electromagnetic field.

3.2 Pressure, inertia and free fall

The local spherical potential-flow comparison, with $\nabla p = -\rho \mathbf{a}_m$ and no additional bulk force, gives [12]

$$F_p = \rho V \mathbf{a}_m, \quad F_a = c_a \rho V (\mathbf{a}_m - \dot{\mathbf{v}}), \quad I_0 = \beta \rho V, \quad \dot{\mathbf{v}} = \frac{1 + c_a}{\beta + c_a} \mathbf{a}_m. \tag{9}$$

The common volume factor cancels. In our hypothesis, the effective coefficient β should be connected to the medium's standing wave. In this comparison it is defined as $I_0/(\rho V)$. Impermeability alone does not fix its value. For a sphere $c_a = 1/2$, and $\beta = 0$ gives $\kappa = 3$, whereas $\beta = 1$ gives unit response. Thus the proposed explanation of equal weight and inertial responses depends on a specific, investigable effective condition.

At $\beta = 1$, without an additional response to velocity difference, radial release from rest at r_0 gives

$$v_r = -\sqrt{2E\left(\frac{1}{r} - \frac{1}{r_0}\right)}, \qquad \left|\frac{v_r}{u_r}\right| = \sqrt{1 - \frac{r}{r_0}}. \tag{10}$$

We show the difference between body and flow speeds without dependence of this ratio on body inertia. For release 1 km above a surface of radius 6371 km, the ratio at the surface is approximately 0.01253. This is a result of the specified model, not a measured speed of the non-material medium.

3.3 Continuity and distinct radial regimes

The stationary profile has $\nabla \cdot \mathbf{u} = -3\sqrt{E/(2r^3)}$. Constant averaged ρ requires $\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = -\rho\sigma$ with $\sigma = 3\sqrt{E/(2r^3)}$. We associate this term with possible formation of new atoms, not entry into existing ones. It is inferred from the profile rather than from an independent formation law. In contrast, incompressible flow around a moving spherical boundary gives $u_r = R^2\dot{R}/r^2$. These regimes must not be identified or mechanically superposed with circular flow [6, 8].

4 Standard results and possible differences

4.1 Agreement already established

In the central regimes examined, at $E = GM$, $\beta = 1$ and $\lambda = 0$, the body acceleration is $-E\mathbf{e}_r/r^2$. Keplerian orbits and radial free fall then agree with standard inverse-square predictions. We explicitly establish this agreement. The standard comparison already cancels the test-body mass and the kilogram unit as shown in Section 1.3. Numerical reproduction of an orbit does not discriminate between two physical interpretations of the same equation.

Our physical proposal adds a medium description, flow around atoms, a medium-phase balance and a common effective response. A difference requires determination of at least one new measurable quantity: a composition-dependent β, an additional coefficient λ, a phase-change law or a geometrical rotor response. Further verification therefore has a specified subject.

4.2 What the software calculates

In the interacting point-source control, the software uses

$$\ddot{\mathbf{r}}_i = \sum_{j \neq i} E_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}. \tag{11}$$

All enabled bodies enter mutual acceleration in the implementation. The stellar parameter sum displayed for a group is a separate diagnostic. Orbital initial states are not forcibly maintained on their initial curves. The calculation uses velocity Verlet, monitors close encounters and provides energy and momentum diagnostics [10]. Parameters of synthetic bodies are specified as experiment inputs. They must not be mistaken for observational determinations for arbitrary objects.

With $E_j = GM_j$, equation (11) is the standard point-source N-body equation. The code tests consequences of the specified law and group replacements. It does not derive interacting media from pictures of flow lines. Prescribed-flow lines, body trajectories and reference orbits have different physical meanings.

4.3 Prospects for multiple systems

Hierarchical presentation, shared translation and local motion allow different scales of a configuration to be studied. Stability requires separate checks: fixed initial conditions, a specified duration, smaller time steps, invariant monitoring and perturbation sensitivity. Visually regular trajectories do not replace these checks.

Known special three-body solutions, including the figure-eight choreography [15], provide useful control problems. The next computational stage may include stability maps, resonance and periodic-configuration searches, and comparisons of full dynamics with group approximations. Parameter changes during an experiment must be recorded as interventions or new initial states. We do not claim a general analytical solution of the three-body problem.

5 Three tasks for independent verification

5.1 Body composition and response coefficient

For two bodies in the local comparison, with equal c_a and $\lambda = 0$,

$$\eta_{AB} = \frac{2(\kappa_A - \kappa_B)}{\kappa_A + \kappa_B}, \qquad \kappa_i = \frac{1 + c_a}{\beta_i + c_a}. \tag{12}$$

This specifies a connection between the hypothesis and differential acceleration. MICROSCOPE measured the acceleration ratio of titanium and platinum test masses and detected no violation of the equivalence principle [13]. Comparison requires deriving β_i from the standing-wave model and translating the prediction into the instrument's actual geometry and frequency channel. **Output:** a quantitative prediction of η with uncertainty. **Completion criterion:** the prediction is fixed before experimental comparison and is compatible or incompatible with the measured bound.

5.2 Gravimetry of rising and falling branches

The possible extension $\dot{\mathbf{v}} = \mathbf{a}_m + \lambda(\mathbf{u} - \mathbf{v})$ introduces a velocity-dependent term. For unit baseline response in a local upward vertical coordinate, write $\ddot{z} = -g_s + \phi\dot{z} + \cdots$, with $\phi_{\text{obs}} = -\lambda + \phi_{\text{known}} + \phi_{\text{other}}$. A slowly varying local background is absorbed into g_s in this branch comparison. For idealized equally spaced distance sampling and quadratic least-squares branch fits, let T be the total trajectory duration and $\tau = T/2$ the duration of each branch. For the branch-weighting method described by Nagornyi, Biolcati and Svitlov [14], the first time moment is $C_1 = 4\tau/7 = 2T/7$. To first order, the magnitude of the contrast due to λ alone is

$$\Delta g_{\uparrow\downarrow} \equiv |g_\uparrow - g_\downarrow| = \frac{4}{7}|\lambda| g T. \tag{13}$$

The two branches have opposite velocity-dependent shifts, so their magnitude difference is $2|\lambda| g C_1$. For equally spaced time sampling the corresponding coefficient is $1/2$, not $4/7$. The sampling rule and fitting procedure therefore belong to the prediction.

For $g = 9.8$ m/s^2, total $T = 0.3039$ s and specified $|\lambda| = 10^{-6}$ s^{-1}, Eq. (13) gives 170.184 microgal. This is an isolated model control signal, not a measured IMGC-02 residual. Real sampling requires the actual weighting function.

We propose collaboration with INRIM researchers on IMGC-02 branch analysis using its processing methodology [14]. **Output:** an empirical bound on λ after separating residual-gas, finite-light-speed and other known effects. **Completion criterion:** one model explains both branches within independently assessed uncertainty. Access to data or an agreed collaboration is not claimed here.

5.3 Rotor geometry and force controls

The third task is to derive the integrated pressure and inertial response for a specified rotor geometry. It requires a predicted force or torque, its speed and orientation dependence, and a control regime in which the proposed effect vanishes. **Output:** a numerical signal and error budget for a defined apparatus. **Completion criterion:** the prediction differs from known mechanical, thermal and electromagnetic contributions and is tested by independent control of them. The direction of satellite drift alone is not an alternative to a discriminating test of this kind.

These tasks define the request for research support. The sequence is derivation, a fixed prediction and independent measurement. Schedule and budget should be agreed with laboratories according to actual access to data and equipment.

6 Conclusions and invitation to verification

6.1 What the series establishes

One physical process, one organizing formula. We propose that the pressure-driven motion of the non-material medium toward aggregates of atoms and around impermeable atoms is the common physical basis of weight and inertia, and that $E = \omega^2 R^3$ is its expression. We show that the formula organizes one computational route from orbital parameters to acceleration, controlled multiple systems and body response. The series establishes the concrete links: a source-group bound with coefficient 3, the inverse-square acceleration $E/r^2 = \lambda_{\mathrm{m}} U$ as the medium's own acceleration, the approach of a released body to the flow as $\sqrt{1 - r/r_0}$, and unit response $\kappa = 1$ for weight and inertia at $\beta = 1$. We do not replace this picture with a permeable atomic sink.

The results have defined verification conditions. We derive the acceleration of a specified radial profile, its continuity balance, a group-replacement bound and a conditional gravimetric branch contrast. The software permits reproduction of specified calculations and exploration of configurations. Agreement of central predictions with standard results identifies where further searches for differences belong.

Invitation to researchers and supporters. We invite examination of this programme through its formulas, data and executable code, reproduction of its results and reports of specific discrepancies. The connected results make the worldview formulation of Kepler's third law a concrete subject for independent examination. We call for its systematic use and verification across these tasks. We invite metrology, physics and astronomy groups to independent verification and support for the three specified tasks. The contribution set out in this register and the companion studies is claimed as the contribution of the Ew2R3.org programme, with its stated bibliographic history.

6.2 Next step

The effective inertia of the atomic standing wave, the medium-balance law and quantitative differences from standard predictions remain to be derived. These tasks determine physical confirmation of the proposed interpretation. Different motion directions and complex geometry require extension of the local spherical comparison. Computational checks do not replace new independent measurements.

Declarations

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Data and code availability

The previous visualization release is archived on Zenodo [9]. Shared calculations, input data, synthetic controls and the series visualization are described in the Ew2R3 software release [10]. Software website: Ew2R3.org. This article creates no new observational data. Its numerical examples can be reproduced from the stated formulas and inputs.

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